Ye Wang

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Employment

Associate Professor, School of Artificial Intelligence, CQUPT — 2026–present
Assistant Professor, School of Computer Science and Technology, CQUPT — 2020–2025
Project Manager (on secondment), China Scholarship Council, Beijing — 2020–2021
Research Scientist, Samsung Research America, United States — 2019

Education

Ph.D. in Computer Engineering, Texas A&M University, United States — 2014–2019
M.S. in Electrical Engineering, The University of Texas at Dallas, United States — 2012–2014
B.Eng. in Microelectronics, Chongqing University of Posts and Telecommunications, China — 2007–2011

Academic Profile

My research develops **human-centered cognitive AI** that can understand individuals and groups, reflect diverse human judgments, and reason reliably from incomplete, uncertain, or conflicting evidence. I pursue this goal through three connected directions.

Research Agenda

- **Human and Collective Understanding:** model emotions, intentions, preferences, human attributes, activities, social narratives, and collective behavior from multimodal and behavioral data.
- **Human–AI Alignment:** represent plural and sometimes conflicting human judgments in model learning and generation rather than reducing them to a single target or average response.
- **Evidence-Grounded Reasoning:** build systems that remain reliable under partial observability, hallucinations, uncertainty, long-tailed conditions, distribution shifts, and competing sources of evidence.

Ongoing Research Projects

Human Difference and Individual Dynamics — *2024–2029*
Overview. This project examines how AI can represent structured differences in human judgment and model individual trajectories over time. It connects plural preference modeling with personalized, identity-preserving facial aging, moving beyond average scores and average patterns of change.
Progress. Our 2026 Spotlight paper at the *International Conference on Machine Learning* (ICML), *When Attributes Disagree*, shows that different aesthetic attributes can produce systematic optimization conflicts and develops a conflict-aware framework for coordinating them. Our 2026 paper in *IEEE Transactions on Image Processing* (TIP), *Hierarchical Causal Learning for Face Age Synthesis*, models dependencies among age-related facial attributes while preserving identity and attribute consistency. Current work extends these foundations toward context-sensitive preference alignment and personalized facial-aging trajectories.

Reasoning Beyond Visible Evidence — *2025–2030*
Overview. This project studies how AI should reason when safety-critical actors, objects, or causal conditions are not directly observable. Its main application is autonomous driving under occlusion, where a system must infer hidden road users and emerging risks from visible agents, environmental context, and other indirect evidence.

Progress. This line builds on three complementary foundations: our 2026 work in *IEEE Transactions on Circuits and Systems for Video Technology* (TCSVT) on grounding model correction in visual evidence, our 2026 work in *IEEE Transactions on Audio, Speech, and Language Processing* (TASLP) on uncertainty-aware learning for difficult minority cases, and our Best Student Paper at the 2026 *International Conference on Neural Computing for Advanced Applications* (NCAA) on generating rare safety-critical driving scenarios. Current research treats visible agents as reliability-dependent proxies for hidden risks and studies mixed human–autonomous traffic, cascading occlusions, and risk-aware planning before a hidden threat becomes visible.

Latent Patterns in Social and Multi-Agent Systems *2025–2030*
Overview. This project investigates how local expressions, actions, and interactions give rise to collective narratives, strategies, and behavioral patterns. It seeks to identify and explain latent positions, intentions, and organizing structures across societies, events, teams, and competitive settings.
Progress. In multilingual news, we are developing an event-aligned corpus and explanatory framework for differences in factual emphasis, stance, responsibility attribution, legitimacy judgment, and value framing, and testing whether these differences can be explained and predicted across events. The broader multi-agent direction builds on our 2025 work in *IEEE Transactions on Affective Computing* (TAFFC) on relational interaction modeling and our 2025 work in *Information Processing & Management* on hierarchical multimodal interaction in football. Ongoing sports research explores latent strategies, player coordination, and changes in match dynamics.

Research Funding

Principal Investigator

National Natural Science Foundation of China, Young Scientists Fund 2024–2026
Chongqing Energy Big Data Center, Research Program 2024–2026
Chongqing Returned Overseas Scholars Support Program, 2022–2023
Science and Technology Research Program of Chongqing Education Commission, 2021–2024

Research Awards

Spotlight Paper, ICML, Top 2.2% 2026
Gold Reviewer, ICML, 2026
Best Student Paper Award, NCAA, 2026
Best Conference Paper, International Conference on Brain Informatics, 2024
Best Paper Candidate, IEEE International Symposium on Product Compliance Engineering - Asia, 2023

Teaching

Teaching Honors

- **2025** — Course Leader, *Data Structures for International Students*; designated a Chongqing First-Class Undergraduate Course.
- **2024** — Award for Outstanding Contribution to International Student Education in Chongqing.
- **2023** — Outstanding Advisor, RoboCom Robot Developer Competition, National Finals.

Courses at CQUPT

- **Natural Language Processing** *(Fall 2024; Fall 2025)*
- **AI Literacy and Practice: From Fundamentals to DeepSeek Applications** *(Spring 2025; Fall 2025)*
- **Algorithm Analysis and Design for International Students** *(Fall 2025)*
- **Data Structures for International Students** *(Spring 2023; Spring 2024; Spring 2025)*
- **Machine Learning Models and Algorithms** *(Fall 2022; Spring 2023; Fall 2024)*
- **Principles of Artificial Intelligence** *(Fall 2020)*
- **Data Mining** *(Spring 2022; Fall 2022; Fall 2024)*
- **Big Data Analytics and Mining** *(Fall 2023; Fall 2024)*

Teaching Assistant at Texas A&M University

- **ECEN 651: Microprogrammed Control of Digital Systems** — *Fall 2014; Spring 2015*.
- **ECEN 350: Computer Architecture** — *Spring 2015; Fall 2017–Spring 2019*.
- **ECEN 214: Electrical Circuit Theory** — *Spring 2016*.

Students Mentoring

- **2025** — National Third Prize, 19th Challenge Cup National College Students' Extracurricular Academic Science and Technology Contest; CQUPT Outstanding Undergraduate Thesis Awards.
- **2024** — National Third Prize, RoboCom Robot Developer Competition; Outstanding Undergraduate Thesis of Chongqing; National Undergraduate Innovation and Entrepreneurship Training Program rated Outstanding upon completion.
- **2023** — Second Prize, Mathematical Contest in Modeling; CQUPT Outstanding Undergraduate Thesis Award; National First Prize, Math China Mathematical Modeling Competition; National Second Prize, China College Students' Service Outsourcing Innovation and Entrepreneurship Competition.

Academic Service

Editorial and Professional Society Service

- Young Editorial Board Member, *CAAI Transactions on Intelligent Systems*.
- Committee Member, CAAI Technical Committee on Granular Computing and Knowledge Discovery.
- Corresponding Member, CAAI Technical Committee on Artificial Intelligence Foundations.

Session Chair, Mesoscopic Cognitive Machine Learning Workshop, IJCRS 2026

Competition Chair, NCAA 2023–2026

Web Chair, IEEE International Conference on Medical Artificial Intelligence 2024

Conference Reviewer: ICML (2026); AAAI (2026); COLM (2024–2026); ACM Multimedia (2026); ACL Rolling Review (2025–2026); ECAI (2022–2025); IJCAI (2023–2024).

Journal Reviewer: TIP; TASLP; TCSVT; TKDE; TMM; TAFFC.

Invited Talks and Media

Invited Talk, "Visual Intelligence through Multi-Granularity Cognitive Computing," CCF@U (No. 1323), Sichuan University 2025

Invited Talk, "Generative Modeling through Latent-Space Disentanglement," Doctoral Forum, Yan'an University 2023

Invited Talks, Summer Training Program on Machine Learning, National Center for Applied Mathematics, Chongqing 2023

Invited Talks, "Generative AI in Education: Opportunities and Challenges," Chongqing Nankai Secondary School 2023

Media Interview, *People's Education* 2023

Media Interview, "ChatGPT Is Here: Should We Be Anxious?", *Chongqing Daily* 2023

Books and Monographs

[B1] **Ye Wang**, Hong Yu, and Guoyin Wang. *Multi-Granularity Cognitive Representation.* Science Press, forthcoming, 2026.

Representative Publications

Journal Papers

[J12] **Ye Wang**, Pan Sun, Xuyang Zhou, Lifeng Shen, Jiaxu Leng, Guoyin Wang, and Hong Yu*. "Hierarchical Causal Learning for Face Age Synthesis." *IEEE Transactions on Image Processing*, 2026.

[J11] Guangyong He, Li Liu*, Youmin Zhang, **Ye Wang**, Qun Liu, and Guoyin Wang. "Enhancing Graph Neural Network Explainers Using a Distribution Shift Consistency-Guided Generator." *IEEE Transactions on Knowledge and Data Engineering*, 2026.

[J10] **Ye Wang**, Zixuan Wu, Lifeng Shen, Jiang Xie, Xiaoling Wang, Hong Yu*, and Guoyin Wang. "Mastering the Minority: An Uncertainty-Guided Multi-Expert Framework for Challenging-Tailed Sequence Learning." *IEEE Transactions on Audio, Speech, and Language Processing*, 2026.

[J9] **Ye Wang**, Jiancheng Zhou, Qun Liu*, Feng Hu, and Guoyin Wang. "Visual Evidence-Aware Object Hallucination Rectification in LLM-Based Video Captioning." *IEEE Transactions on Circuits and Systems for Video Technology*, 2026.

[J8] **Ye Wang**, Qingyan Wang, Hong Yu, Jiang Xie, Feng Hu, Xiaoling Wang, and Dajiang Lei*. "GCFA: Generative Class Feature Fusion with Agent Attention for Medical Text Classification." *Information Fusion*, 2026.

[J7] **Ye Wang**, Xinyang Li, Hong Yu, Feng Hu, Guoyin Wang*, and Dajiang Lei*. "Continuous Entity Reasoning for Multi-Turn Medical Dialogue Generation." *IEEE Transactions on Consumer Electronics*, 2025.

[J6] **Ye Wang**, Qi Wei, Hong Yu, Guoyin Wang, Chunmeng Shi, and Dajiang Lei*. "Cross-Interaction of Chinese Character Structures and Boundary Features for Improving Clinical Named Entity Recognition." *IEEE Journal of Biomedical and Health Informatics*, 2025.

[J5] Xuyang Zhou, **Ye Wang***, Fei Tao, Hong Yu, and Qun Liu. "Hierarchical Chat-Based Strategies with MLLMs for Spatio-Temporal Action Detection." *Information Processing & Management*, 2025.

[J4] **Ye Wang**, Wei Zhang, Ke Liu*, Wei Wu, Feng Hu, Hong Yu, and Guoyin Wang. "Dynamic Emotion-Dependent Network with Relational Subgraph Interaction for Multimodal Emotion Recognition." *IEEE Transactions on Affective Computing*, 2025.

[J3] **Ye Wang**, Zheng Wang, Hong Yu, Guoyin Wang*, and Dajiang Lei*. "The Interactive Fusion of Characters and Lexical Information for Chinese Named Entity Recognition." *Artificial Intelligence Review*, 2024.

[J2] **Ye Wang**, Qianmengke Zhao, Qun Liu*, Guoyin Wang, Hong Yu, Li Liu, and Jiaxu Leng. "KDDGAN: Knowledge-Guided Explicit Feature Disentanglement for Facial Attribute Editing." *IEEE Transactions on Consumer Electronics*, 2024.

[J1] **Ye Wang**, Xinxiang Zhang, Mi Lu, Han Wang, and Yoonsuck Choe. "Attention Augmentation with Multi-Residual in Bidirectional LSTM." *Neurocomputing*, 2020.

Conference Papers

[C7] **Ye Wang**, Maocai Dai, Jiang Xie, Xiuli Bi, Fei Tao, Xiao Li, and Hong Yu. "When Attributes Disagree: Gradient Conflict in Image Aesthetic Assessment." *International Conference on Machine Learning (ICML)*, 2026. **Spotlight, Top 2.2%.**

[C6] Tingting Lei, Yifan Zhu, Runxi Zhang, Feng Hu, Hong Yu, and **Ye Wang**. "NLG-Gen: Natural Language-Guided Generation of Long-Tail Critical Scenarios for Autonomous Driving." *International Conference on Neural Computing for Advanced Applications (NCAA)*, 2026. **Best Student Paper Award.**

[C5] Wenbo Wu, Qingyi Si, Xiurui Pan, **Ye Wang**, and Jie Zhang. "LouisKV: Efficient KV Cache Retrieval for Long Input-Output Sequences." *International Conference on Learning Representations (ICLR)*, 2026.

[C4] Yan Xian, Hong Yu, **Ye Wang**, and Guoyin Wang. "A Novel Class Incremental Learning Method via Multi-Granularity Balance Inspired by Human Granular Cognition Mechanism." *International Conference on Brain Informatics*, 2024. **Best Conference Paper.**

[C3] Xinqiang Jiang, Yingnan Geng, Yinzhou Xiong, Fei Tao, and **Ye Wang**. "A Privacy-Aware Framework for Assessing and Recommending Short Video Advertisement." *IEEE International Symposium on Product Compliance Engineering - Asia (ISPCE-ASIA)*, 2023. **Best Paper Candidate.**

[C2] Jiaxu Leng and **Ye Wang**. "RCNet: Recurrent Collaboration Network Guided by Facial Priors for Face Super-Resolution." *IEEE International Conference on Multimedia and Expo (ICME)*, 2022.

[C1] **Ye Wang**, Han Wang, Xinxiang Zhang, Theodora Chaspari, Yoonsuck Choe, and Mi Lu. "An Attention-Aware Bidirectional Multi-Residual Recurrent Neural Network: A Study on Short-Term Text Classification." *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2019.

Additional Collaborative Publications

[A27] Zhuoyi Yu, Guanmeng Xian, Qianmengke Zhao, Hong Yu, Xiao Li, and **Ye Wang***. "Multi-Task Learning with Hierarchical Feature Disentanglement for Low-Exposure Facial Recognition in Smart Cockpits." *Neural Computing and Applications*, 2026.

[A26] Yan Xian, Hong Yu, **Ye Wang**, and Guoyin Wang. "Exploring Multi-Granularity Balance Strategy for Class Incremental Learning via Three-Way Granular Computing." *Brain Informatics*, 2025.

[A25] Chenrui Ma, Yuxin Ding, Haijun Zhang, Jianghong Ma, Zhao Zhang, **Ye Wang**, and Fei Tao. "Fine-Grained Offline Ranking for Short Video Advertisements CTR via Visual- and Audio-Based Features." *IEEE Transactions on Consumer Electronics*, 2025.

[A24] Zixuan Wu, **Ye Wang**, Lifeng Shen, Feng Hu, and Hong Yu. "Leveraging Uncertainty for Depth-Aware Hierarchical Text Classification." *Computers, Materials & Continua*, 2024.

[A23] Hong Yu, Jinxuan Tang, Zhihan Peng, and **Ye Wang**. "Relation Correlations-Aware Graph Convolutional Network with Text-Enhanced for Knowledge Graph Embedding." *International Journal of Machine Learning and Cybernetics*, 2024.

[A22] **Ye Wang**, Daitianxia Li, Qun Liu, Li Liu, and Guoyin Wang. "Self-Supervised Modal Optimization Transformer for Image Captioning." *Neural Computing and Applications*, 2024.

[A21] Youmin Zhang, Lei Sun, **Ye Wang***, Qun Liu, and Li Liu. "One-Shot Knowledge Graph Completion Based on Disentangled Representation Learning." *Neural Computing and Applications*, 2024.

[A20] Kangjia He, Li Liu, Youmin Zhang, **Ye Wang**, Qun Liu, and Guoyin Wang. "Learning Counterfactual Explanation of Graph Neural Networks via Generative Flow Network." *IEEE Transactions on Artificial Intelligence*, 2024.

[A19] Pan Sun, Hong Yu, Jiang Xie, Jiaxu Leng, and **Ye Wang**. "Explicit Facial Attribute Disentanglement for Hierarchical Relationships Detection." *International Conference on Neural Computing for Advanced Applications (NCAA)*, 2024.

[A18] Shijie Sun, Hanyue Liu, **Ye Wang**, and Hong Yu. "Lung Cancer Risk Prediction Model Trained with Multi-Source Data."

International Joint Conference on Rough Sets (IJCRS), 2024.

[A17] Yali Wang, **Ye Wang**, Hong Yu, and Ke Liu. “Steering Angle Prediction Based on Travelable Regions and Bio-Inspired Neural Circuit Policy.” *International Joint Conference on Rough Sets (IJCRS)*, 2024.

[A16] Xu Xiang, Hong Yu, **Ye Wang**, and Guoyin Wang. “Stable Local Interpretable Model-Agnostic Explanations Based on a Variational Autoencoder.” *Applied Intelligence*, 2023.

[A15] Jiaxu Leng, Haitao Wang, Xinbo Gao, Yan Zhang, **Ye Wang**, and Mengjingcheng Mo. “Where to Look: Multi-Granularity Occlusion Aware for Video Person Re-Identification.” *Neurocomputing*, 2023.

[A14] **Ye Wang**, Jingbo Liao, Hong Yu, Guoyin Wang, Xiaoxia Zhang, and Li Liu. “Interpreting Open-Domain Dialogue Generation by Disentangling Latent Feature Representations.” *Neural Computing and Applications*, 2023.

[A13] Zhuoyi Yu, **Ye Wang**, and Dajiang Lei. “Research on Chinese Diabetes Question Classification with the Integration of Different BERT Models.” *International Conference on Neural Computing for Advanced Applications (NCAA)*, 2023.

[A12] Qingyan Wang, **Ye Wang**, and Dajiang Lei. “SFDA: Chinese Diabetic Text Classification Based on Sentence Feature Level Data Augmentation.” *International Conference on Neural Computing for Advanced Applications (NCAA)*, 2023.

[A11] Daitianxia Li, **Ye Wang**, and Qun Liu. “Adaptive Multi-Granularity Aggregation Transformer for Image Captioning.” *International Joint Conference on Rough Sets (IJCRS)*, 2023.

[A10] Xiaoxia Zhang, Lu Chen, **Ye Wang**, and Guoyin Wang. “Improving Incremental Nonnegative Matrix Factorization Method for Recommendations Based on Three-Way Decision Making.” *Cognitive Computation*, 2022.

[A9] Li Liu, Yong Du, **Ye Wang**, William K. Cheung, Youmin Zhang, Qun Liu, and Guoyin Wang. “LRP2A: Layer-Wise Relevance Propagation Based Adversarial Attacking for Graph Neural Networks.” *Knowledge-Based Systems*, 2022.

[A8] **Ye Wang**, Jingbo Liao, Hong Yu, and Jiaxu Leng. “Semantic-Aware Conditional Variational Autoencoder for One-to-Many Dialogue Generation.” *Neural Computing and Applications*, 2022.

[A7] Li Liu, Chongyang Wang, Youmin Zhang, **Ye Wang**, Qun Liu, and Guoyin Wang. “Denoise Network Structure for User Alignment Across Networks via Graph Structure Learning.” *International Conference on Data Mining and Big Data (DMBD)*, 2022.

[A6] Jingbo Liao, Hong Yu, Qi Cheng, and **Ye Wang**. “Context-Aware Multi-Feature Fusion for Open-Domain Dialogue Generation.” *IEEE International Conference on Industrial Technology (ICIT)*, 2022.

[A5] Jiaxu Leng, Yihui Ren, Wen Jiang, Xiaoding Sun, and **Ye Wang**. “Realize Your Surroundings: Exploiting Context Information for Small Object Detection.” *Neurocomputing*, 2021.

[A4] Shan Jin, Jiafan Wang, **Ye Wang**, and Zhengguang Zheng. “A New Method for Sparse Signals Reconstruction in Fusion Center with Incomplete Measurements.” *IEEE International Conference on Communications (ICC)*, 2019.

[A3] Xinxiang Zhang, **Ye Wang**, Han Wang, Yue Zhang, Hao Wu, and Mingbo Zhao. “Bend Detection of Bridge Chords in UAV Images via Region-Based Deep Semantic Segmentation Network.” *IEEE International Conference on Industrial Informatics (INDIN)*, 2019.

[A2] Han Wang, **Ye Wang**, Xinxiang Zhang, Mi Lu, Yoonsuck Choe, and Jingjing Cao. “English Out-of-Vocabulary Lexical Evaluation Task.” *IEEE International Conference on Industrial Informatics (INDIN)*, 2019.

[A1] **Ye Wang**, Yuanjiang Huang, Wu Zheng, Zhi Zhou, Debin Liu, and Mi Lu. “Combining Convolutional Neural Network and Self-Adaptive Algorithm to Defeat Synthetic Multi-Digit Text-Based CAPTCHA.” *IEEE International Conference on Industrial Technology (ICIT)*, 2017.